

The Fabric of Sustainability: HCI and the Fight Against Textile Waste

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Abstract

The 'Perpetual Pigments' project originated from a discovery by researchers at Deakin University's Institute for Frontier Materials, who extracted pigments from discarded natural fibre textiles. Institute for Frontier Materials collaborated with design academics to explore applications of this breakthrough using design thinking and Indigenous Design Charters. Prominent Australian First Nation artists tested the recycled pigments through painting, drawing, and screen printing. The project's success relied on ubiquitous computing and digital design techniques. This initiative, inspired by the circular economy and research through design, merges science, design, and art to address cultural, economic, and environmental challenges. By adopting a circular economy mindset, textile waste can be repurposed, reducing waste and conserving resources. This approach highlights the principles of functionality, sustainability, and storytelling. The exhibition and findings are valuable to design and Human Computer Interaction researchers and industry representatives interested in addressing textile waste through successful circular economy projects.

CCS Concepts

• Human-centered computing → Field studies.

Keywords

Circular economy; Design research; Information visualization; Exhibition practice; Recycled fibres

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1 Introduction

In an era where environmental challenges are becoming increasingly dire, the issue of textile waste stands out as a significant contributor to global pollution. Every year, millions of tons of textiles are discarded, clogging landfills, and polluting ecosystems. Yet,

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within this crisis lies an opportunity to rethink and redesign our approach to textiles, turning waste into a valuable resource. This notion is eloquently captured in the words of design scholars Page and Memmott, alongside visionary thinker R. Buckminster Fuller: '...all of the design principles - functionality, sustainability, storytelling – are interrelated and work together to define Indigenous design. More importantly, they are a blueprint for the creation of new artefacts, new ceremonial grounds and new architecture and engineering to produce a dynamic Australian design that will last perhaps another 65,000 years' [5].

'Pollution is nothing but the resources we are not harvesting. We allow them to disperse because we've been ignorant of their value' [7].

This combined perspective offers insight into how we can address the issue of textile waste. The principles of Indigenous design and HCI offer a profound blueprint, merging functionality, sustainability, and storytelling to create a dynamic Australian design that could endure for another 65,000 years. HCI, Human-Computer Interaction, is a branch of computer science that is designed to achieve an optimum user interface between humans and computers [8]. Indigenous design principles emphasize the interconnectedness of functionality, sustainability, and storytelling, providing a holistic approach to creation and innovation. These principles are not just about preserving traditions but about laying the groundwork for future generations to develop new artefacts, spaces, and systems that are inherently sustainable and resilient.

At the heart of this philosophy is the recognition that waste, particularly textile waste, is a resource waiting to be harnessed. Buckminster Fuller's observation highlights a critical flaw in our current systems: we allow valuable materials to disperse into the environment because we fail to recognize their potential. By rethinking waste as a resource, we can develop innovative methods to recycle and repurpose textiles, reducing the burden on landfills and the environment.

One promising approach is the extraction of pigments from discarded textiles made of natural fibres. This pioneering method, developed by researchers at the Institute for Frontier Materials (IFM) at Deakin University, showcases how waste can be transformed into valuable resources for new designs. By collaborating with design academics and Indigenous artists, the IFM project illustrates the power of combining scientific innovation with traditional knowledge and design methods thinking.

Digital and communication design techniques play a crucial role in this transformation. Tools for 3D modelling, and digital fabric printing, enable designers to visualise and experiment with recycled



Figure 1: Figure 1: Recycled textile waste as pigment in brown, red and yellow, sprinkled to canvas. Image copyright Tonya A. Meyrick

materials in new and exciting ways. These technologies facilitate collaboration, allowing diverse teams to work together seamlessly, sharing ideas and feedback in real time. Moreover, digital platforms for web design and social media expand the reach of these innovations, engaging a broader audience and fostering a culture of sustainability.

Addressing the urgent issue of textile waste necessitates adopting a circular economy mindset, where materials are continuously repurposed and reused. This approach not only mitigates waste but also conserves resources and reduces environmental impact. Embracing principles of functionality, sustainability, and storytelling, and recognizing the value in discarded materials, can establish a dynamic and enduring design legacy.

The call to action is clear: we must innovate, collaborate, and educate to reduce textile waste. By doing so, we honour the wisdom of Indigenous design principles and the visionary insights of thinkers like Buckminster Fuller, paving the way for a sustainable future that values and utilizes every resource at our disposal.

One of the world's largest polluters, the textile industry stands at a crossroads. Each year, millions of tons of textiles end up in landfills, contributing significantly to environmental degradation. The challenge is not just in the sheer volume of waste but in the systemic neglect of valuable resources embedded within these discarded fabrics. Today, we must heed Fuller's insight and recognise that what we see as waste is, in fact, a reservoir of untapped potential.

Indigenous design principles, emphasising sustainability and resourcefulness, offer a powerful response to this crisis. Traditional practices demonstrate a deep understanding of materials and a commitment to longevity and circularity. By integrating these principles, we can transform textile waste into valuable resources, creating new materials, products, and ecosystems that reflect a sustainable future.

2 The Project - Looking Forward

Our work expands upon the Perpetual Pigments project [3], which focused on exploring applications of this innovation using design thinking and HCI, technology, art, and fashion, and incorporated Indigenous Knowledge protocols through the *Indigenous Design Charter's* best practices. In the current paper and subsequent exhibition, *The Fabric of Sustainability, HCI and the Fight Against*

Textile Waste the researchers provide advance this story for new audiences to rethink approaches to discarded textiles, turning a forgotten waste into a valuable resource and to consider how HCI can contribute to this important area.

Achieved in a dynamic exhibition design that celebrates this story, we explore components of the circular economy through a procedure of reuse and regeneration. We physically and digitally remix design artefacts, and information graphics across physically wearable and digital immersive properties. This is accomplished to engage in a critical discourse focused on the vital call to action, and to consider that which we have not been exploiting. We open a window to view process and procedure of how HCI contributes to solutions in this important space.

Our exhibition practice is a hybrid and live space, an encounter to leverage our previous field studies and foster creative risk-taking, experimentation and connection to expand upon our initial premise. We visually unpack and illustrate concepts of a circular economic system based on the recycling and reintegration of fabric waste, across the physical site of the gallery through projected and printed information visualizations coupled with expanded design processes.

Many aspects of HCI reduce somatic engagement, however, we counter this to increase physical engagement with audiences. Actual piles of discarded textile waste are sculpturally offered for contemplation with moving image documentary footage projected onto piles in an emotive and engaging way. Screen-printed recycled waste pigment designs on fabrics are presented whereby audiences may materially participate in feeling and touching ground textile waste in this formula.

An oversized assemblage of posters convenes behind the pile of discarded textile waste to highlight the characteristics of creative processes, with aspects of 3D modelling and visualizations of the circular economy explored. A meditation on design process is offered here, inviting a deeper consideration of space and resources. Accurate digital replicas of workflows and outcomes of quick experiments of different designs and patterns are shown, to demonstrate how HCI capability may significantly accelerate projects in the circular economy.

We actively engage in the notion of an expanded design process or a 'pulling back of the curtain' on the practices of reusing textile waste to highlight the contributions of design and ubiquitous computing technology to this cause. Allowing audiences to observe the gallery location as a site-specific opportunity for the installation that is both physically present yet temporarily fleeting.

Our approach to exhibition practice positions the gallery as a boundary for the audience to visually and conceptually engage the experience first-hand. We aim to celebrate the potential of digital technologies to advance the important call to action. Our objective in this work is to unpack human computing design processes and present both a view to the past and a look to the future, seeking new collaborations, and applications as well as refining audiences' knowledge construction in the potential to turn the tide on textile waste. Overall, the exhibition offers an unpacking of process and considers the ways HCI may help develop ongoing systems where scientific discovery and application work together to address issues of textile waste ending up in landfill in Australia each year.

3 Methodology

In the Research through Design [9] methodology, designers engage iteratively staging work through cycles of continuously improvement to achieve design solutions through approaches to refinements, results, tests and reviews. Ubiquitous computing technologies harnessed in this project are not new, and neither is the way in which these technologies have been used together. What is novel about this project is – by engaging contemporary issues of sustainability and a strong motivation to find solutions for these issues to ubiquitous computing, new solutions for sustainability can be found. This project is a rich example of how this might operate.

In *The Fabric of Sustainability: HCI and the Fight Against Textile Waste* we approach the exhibition as an active site, in that fundamentally the capacity to enable this circular economy project is only made possible due to computing technologies. We aim to provoke a collective reflection on the digital mediated processes and design outcomes that serves as a conduit for knowledge dissemination, emphasizing design's role as a conciliator of understanding.

Drawing upon numerous technological approaches the exhibition engages aspects of the production chain through digital engagement to promote and make visible Circular Economy Models. The following digital and communication design techniques were iteratively engaged in developing the exhibition to refine ideas and achieve results. These techniques include:

3.1 1. Digital Fabric Printing:

This enabled the precise and consistent application of recycled fibre pigments. Advanced digital printing techniques were used to test and apply the recycled fibre pigments on various textiles. This method ensured precision and consistency in applying pigments, crucial for evaluating their performance. Digital fabric printing also allowed for quick iterations and modifications, enabling experimentation with different designs and patterns efficiently.

3.2 2. Digital Design Software:

Extensively used to create visual communication materials for informational content, including digital exhibition displays. In developing digital assets, the researchers ensured a cohesive visual identity and effective communication of the project's overarching goals. Multimedia techniques, including videos and animations, explore dynamic and engaging content that highlights the project's process, findings, and significance.

By integrating digital techniques, the Fabric of Sustainability project accelerates its research goals and efficiently meets project objectives in our call to action. This approach expands the possibilities of modern design practices, demonstrating how digital innovation addresses critical issues such as textile waste. By leveraging these digital and communication design techniques, the exhibition not only expanded its goals effectively but also broadened the scope of the project. This underscores the significant impact of digital innovation in contemporary design, fostering greater collaboration, precision, and engagement. Ultimately, enriching project outcomes and underscoring the potential of ubiquitous computing coupled with sustainable design to tackle global challenges.

4 Findings

The application of HCI to support environmental sustainability and the creative application of recycled textile fibers as a circular economy success story results in several outcomes effectively demonstrating our research to address textile waste. This underscores the urgency of addressing textile waste and highlights how design practices and HCI technologies can pave the way for a sustainable future. In this project, Human-Computer Interaction contributed to promoting and achieving environmental sustainability and responsible textile management through the circular economy. There are several ways that HCI may be harnessed for this purpose with the following areas providing a rich start:

4.1 Designing Persuasive Technologies

HCI researchers can design persuasive technologies encouraging pro-environmental behaviours, such as proper waste sorting, recycling, and reducing consumption. These technologies can leverage techniques like gamification, social influence, and feedback mechanisms to motivate sustainable practices [2, 4].

4.2 Facilitating Citizen Engagement

HCI can develop platforms and tools that facilitate citizen engagement in environmental initiatives, enabling collaboration, knowledge sharing, and collective action towards sustainable waste management practices [6].

4.3 Enhancing Waste Management Systems

HCI can play a role in enhancing existing waste management systems by developing technologies that optimize collection routes, automate sorting processes, and provide real-time data for better decision-making and resource allocation [8].

4.4 Promoting Circular Economy Models

HCI researchers can contribute to the design of digital platforms and services that support circular economy models, enabling the reuse, repair, and recycling of products and materials, thereby reducing waste generation [8].

4.5 Addressing Legal and Regulatory Aspects

Human-Computer Interaction (HCI) research should integrate legal and regulatory frameworks to address the systemic conditions and constraints of sustainable waste management. This involves understanding the laws and policies governing waste management, which influence how waste is collected, processed, and disposed of. By incorporating these legal considerations into the design process, HCI researchers can ensure their innovations are compliant and promote sustainable practices. This approach includes thorough research of relevant laws, collaboration with legal experts, and integrating legal requirements into design specifications and testing. Such integration ensures that the developed systems are both effective and legally sound, supporting sustainable waste management practices [1].

4.6 Fostering Environmental Stewardship

HCI can play a role in fostering environmental stewardship by designing technologies that support the coordination and capacity-building of various actors involved in waste management, including individuals, communities, organizations, and governing bodies [6] (Rossitto, Tholander, Comber, Jacobsson, 2022). By adopting a holistic and relational approach that considers the complex socio-technical and ecological contexts of waste management, HCI can contribute to the development of innovative solutions that promote environmental sustainability and responsible resource utilisation [1, 6].

5 Conclusion

In an era where environmental challenges are becoming increasingly dire, the issue of textile waste stands out as a significant contributor to global pollution. Yet, within this crisis lies an opportunity to rethink and redesign our approach to textiles, turning waste into a valuable resource. The path to a sustainable future lies in our ability to rethink and redesign our relationship with textiles. By embracing human centered design with the interrelated principles of functionality, sustainability, and storytelling, and by harnessing the wisdom of past innovators like R. Buckminster Fuller, we can transform textile waste into a valuable resource. Let us act now, inspired by the enduring legacy of Indigenous design, to create a world where waste is not an end, but a beginning of new possibilities.

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